Supervisor Findings Brief: Fire-Hotspot Reliability in NSW

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Status: Preliminary findings for scope confirmation—not a final literature review or detector-performance claim

Proposed direction

Working title: Trustworthy Multi-Source Fire-Hotspot Reliability and Active-Fire Monitoring: An NSW Case Study

The proposed project focuses on fire-hotspot detection and active-fire monitoring, not fire-spread prediction. The literature and operational systems indicate that multi-sensor detection already exists; a stronger research contribution would investigate whether hotspot confidence is reliable under realistic sensor, event, and time variation.

The recommended initial contribution is a reproducible reference-data and sensor-confidence reliability audit, followed by modelling only if the available labels support it.

What the initial review indicates

- MODIS, VIIRS, Himawari/AHI, Sentinel-2, and Landsat offer complementary spatial resolution, revisit frequency, and latency.
- Australian services such as DEA Hotspots and MyFireWatch already provide operational multi-source hotspot monitoring.
- Australian and international research already includes conventional machine learning, deep segmentation, transformers, temporal learning, sensor fusion, and explainability.
- Applying a complex model or combining sensors is therefore not sufficient originality by itself.
- Important unresolved issues include imperfect reference labels, non-comparable confidence scales, event leakage, false-alarm burden, probability calibration, and generalisation to new fires and future periods.

These are preliminary conclusions. Eleven accessible full texts have been collected, but several papers in the seed matrix still require complete metadata and full-text verification before the literature-gap claim is finalised.

Completed public-data pilot

I tested whether public data could support the project before requesting a private dataset.

Item	Pilot result
Region and period	Greater Blue Mountains area, 1–14 January 2020
DEA hotspot observations	19,849
Temporally overlapping NSW Fire History events	14

Item	Pilot result
Exact polygon-and-time matches	2,878 (14.5%)
Matches after sensor-specific positional buffering	3,385 (17.1%)
Unresolved observations after buffering	16,461

The data sources were publicly accessible and the workflow was reproducible. The scientific limitation is label validity rather than data access.

Key pilot findings

1. Unmatched does not mean false alarm. NSW Fire History polygons describe final event extents rather than point-in-time active flame. An unmatched observation could be a true fire missing from the selected reference, a prescribed burn, non-fire heat, geolocation or temporal uncertainty, or a satellite commission error.
2. Position tolerance explains only part of the mismatch. Sensor-specific buffers increased matches by 507, but more than four-fifths of observations remained unresolved.
3. Confidence values cannot be pooled directly. AHI and AVHRR confidence was fixed at 50 in this sample, while VIIRS and MODIS algorithms used different ranges. Calibration must therefore be sensor- and algorithm-aware.
4. Random row splitting would leak fire events. Most buffered matches—2,983—came from one event, `Stockyard Creek; Little`. Evaluation should hold out complete events and future periods.

This pilot is a data-feasibility and reference-quality audit. It is not an estimate of operational detection accuracy.

Refined research question

How reliable are DEA hotspot confidence indicators across sensors, algorithms, fire events, and time in an NSW case study, and can sensor-aware calibration improve their decision usefulness under held-out-event and future-period evaluation?

If stronger point-in-time labels become available, the question could be extended to compare a temporal model with operational-confidence, logistic-regression, and tree-based baselines.

Proposed next stage

1. Repeat the audit across several independent NSW fires and at least one prescribed-burn period.
2. Add Historical Fire Extent and Severity Mapping as a complementary reference.
3. Use transparent labels: confirmed event, unresolved, and confirmed non-fire, rather than forcing unmatched observations into a false-alarm class.
4. Aggregate repeated observations into event candidates.
5. Compare confidence within sensor and algorithm families.
6. Use held-out-event and chronological validation.
7. Add an advanced temporal or graph model only if it answers a demonstrated limitation better than simpler baselines.

Question for supervision

Does a reference-data and sensor-confidence reliability audit, using the three-class treatment above, constitute a suitable initial research contribution? If stronger labels are necessary, do you know of an archived incident record or verified point-in-time active-fire source that could help resolve the unmatched class?